

Outline

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May 5, 2026

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Improving Probabilistic Diffusion Models...

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Background: Diffusion models are transforming generative AI

- State-of-the-art in images, audio, video, 3D, and text.
- High-fidelity synthesis via reversing a noising process.
- Growing demand for faster, reliable, few-step generation.
- Applications need calibrated likelihoods and diversity.

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Background: How diffusion sampling works

- Forward: add Gaussian noise step-by-step.
- Reverse: denoise using learned score network.
- Denoising uses Gaussian mean and covariance.
- Classic practice: learn mean, fix covariance.

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- Few-step samplers are sensitive to covariance choice.
- Fixed or heuristic covariances hurt diversity and likelihood.
- Prior diagonal learners use biased data targets.
- Full Hessians are too costly at scale.

- Controls uncertainty propagation between steps.
- Affects recall and coverage of data modes.
- Impacts tractable likelihood and calibration.
- Key bottleneck when aggressively reducing steps.

- Exploit analytic optimal covariance from the score's Hessian.
- Learn diagonal via unbiased Optimal Covariance Matching (OCM).
- Lightweight Hessian head, $M=1$ Hutchinson training.
- Improves FID–NLL trade-off, Recall, and efficiency.

- Posterior covariance depends only on score's Hessian.
- Scaled Gaussian convolution yields closed-form expression.
- Diagonal target: function of Hessian diagonal and schedule.
- Principled objective independent of data moments.

- Freeze pretrained score; add diagonal head h_ϕ .
- Hutchinson trick with Rademacher vectors v .
- Match $v \odot H(x) v$ in expectation (unbiased).
- $M=1$ suffices; low-variance in practice.

- Share backbone features (UNet or DiT).
- Parallel to score prediction with minimal overhead.
- Negligible memory; seconds per step unchanged.
- Scales to pixel and latent diffusion models.

- Use $M=1$ Rademacher vector per batch.
- Scale diagonal by noise schedule appropriately.
- Clip covariance at late steps for stability.
- Works with skip-step schedules and latents.

- DDPM: sample next state using learned covariance.
- DDIM: replace delta over x_0 with Gaussian.
- Restores stochasticity and improves diversity.
- Supports skip-to-zero trajectories.

Table: Algorithm 1 and Algorithm 2 sampling procedures (tabular layout as in paper)

Algorithm 1 Sampling procedure from $t \rightarrow t''$ in OCM-DDPM	Algorithm 2 Sampling procedure from $t \rightarrow t''$ in OCM-DDIM
1: Compute $\mu_{t'}(x_t)$ with Equation (15); 2: Compute $\Sigma_{t''}(x_t)$ with Equation (16); 3: Sample $\epsilon \sim \mathcal{N}(0, I)$; 4: Let $x_{t''} \leftarrow \mu_{t''}(x_t) + \Sigma_{t''}(x_t)^{1/2} \epsilon$.	1: Compute $\mu_0(x_t)$ with Equation (18); 2: Compute $\Sigma_0(x_t)$ with Equation (19); 3: Sample $x_0 \sim \mathcal{N}(\mu_0(x_t), \Sigma_0(x_t))$; 4: Let $x_{t''} \leftarrow \sqrt{\bar{\alpha}_{t''}} x_0 + \sqrt{1 - \bar{\alpha}_{t''}} \epsilon$.

- DDIM often deterministic, hurting diversity.
- OCM adds principled stochasticity via covariance.
- Better Recall without sacrificing FID.
- Enables accurate likelihood estimation.

- Datasets: CIFAR10, CelebA64, LSUN Bedroom, ImageNet64, ImageNet256 latent.
- Backbones: UNet-based pixels; DiT for latent ImageNet256.
- Baselines: DDPM, DDIM, Improved DDPM, Analytic/NPR/SN variants.
- Metrics: FID, NLL (bits/dim), Recall, MMD (toy).

Key result: Best FID–NLL trade-off on CIFAR10

Controlled validation on Toy MoG

- OCM attains lower FID at fixed NLL.
- OCM attains lower NLL at fixed FID.
- Consistent across step counts.
- Improved efficiency over baselines.

- OCM achieves lowest covariance error.
- Best MMD under few steps.
- Approaches true-diagonal with far fewer steps.
- Confirms theoretical grounding.

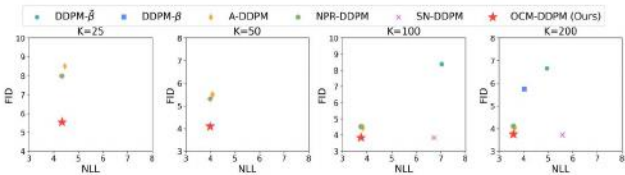


Figure: FID与NLL随采样步数变化的各法对比（CIFAR-10, CS）；本法权衡最佳

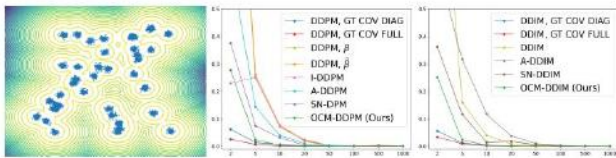


Figure: 协方差估计方法对比: (a)训练数据与真实密度; (b)(c)DDPM/DDIM下MMD评估随总步数变化

- OCM needs fewer timesteps across datasets.
- Remains best after per-step overhead adjustment.
- ImageNet256 latent: 10 steps (5.33 FID).
- Significant wall-clock savings.

Table: Table 4: The least number of timesteps ↓ required to achieve an FID around 6 (with time-cost adjustment).

METHOD	CIFAR10	CELEBA 64X64	LSUN BEDROOM	IMAGENET 256X256
DDPM	90 (6.12)	> 200	130 (6.06)	21 (5.89)
DDIM	30 (5.85)	> 100	BEST FID > 6	11 (5.58)
IMPROVED DDPM	45 (5.96)	MISSING MODEL	90 (6.02)	22 (6.08)
ANALYTIC-DPM	25 (5.81)	55 (5.98)	100 (6.05)	MISSING MODEL
NPR-DPM	1.002 × 23 (5.76)	1.013 × 50 (6.04)	1.021 × 90 (6.01)	MISSING MODEL
SN-DPM	1.005 × 17 (5.81)	1.019 × 22 (5.96)	1.114 × 92 (6.02)	MISSING MODEL
OCM-DPM (OURS)	1.003 × 16 (5.83)	1.015 × 21 (5.94)	1.112 × 90 (6.04)	1.007 × 10 (5.33)

CIFAR10 (CS): visual quality vs steps

- Stable synthesis under aggressive step reduction.
- Maintains fidelity and diversity.
- Visual evidence matches quantitative gains.



Figure: Generated samples vs. sampling steps, CIFAR-10 (CS) (Tab. 5)

- OCM improves NLL notably over isotropic and VLB-based.
- Competitive or best FID across steps.
- Consistent gains with diagonal learning from score.
- Supports earlier FID–NLL frontier result.

Table: Table 1: Comparison of different covariance choices on CIFAR10 (CS). We use Rademacher estimator (M = 100) for the diagonal covariance approximation in OCM-DDPM.

# timesteps	FID ↓				NLL ↓			
	5	10	15	20	5	10	15	20
DDPM, β	58.28	34.76	24.02	19.00	203.29	74.95	44.94	32.20
DDPM, β		254.07	205.31	149.67	109.81	7.33	6.51	6.06
OCM-DDPM	38.88	21.60	13.35	9.75	6.82	4.98	4.62	4.43

CIFAR10 (LS): visual quality vs steps

- Progressive refinement visible across steps.
- OCM preserves quality with fewer steps.
- Artifacts reduced versus baselines qualitatively.
- Corroborates FID trends.



Figure: CIFAR-10 (LS) 不同采样步数的生成样本 (见表5)

Likelihood: ImageNet 64×64 upper bounds

- OCM achieves lowest NLL across steps.
- Improves density modeling over NPR/SN.
- Closes gap to long-run samplers faster.
- Better quality–likelihood trade-off overall.

Table: Table 10: Upper bound on the negative log-likelihood (bits/dim) on the ImageNet 64×64 dataset.

# TIMESTEPS K	25	50	100	200	400	4000
I-DDPM	18.91	8.46	5.27	4.24	3.86	3.57
NPR-DDPM	4.66	4.22	3.96	3.80	3.71	3.60
SN-DDPM	4.56	4.18	3.95	3.80	3.71	3.63
OCM-DDPM	4.45	4.15	3.93	3.79	3.70	3.59

ImageNet 256×256: metrics across CFG (10 steps)

- Best FID at CFG=2.0 with OCM-DDIM.
- Recall improved across CFG values.
- Quality and diversity simultaneously improved.
- Strong results with only 10 steps.

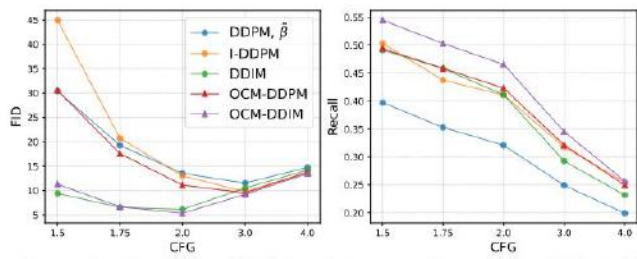


Figure: DiT on ImageNet 256×256: 10-step sampling with varying CFG

ImageNet 256×256: qualitative across CFG

- OCM sustains diversity under stronger guidance.
- Sharper details with maintained variety.
- Better mode coverage than baselines.
- Visuals align with Recall gains.



Figure: ImageNet 256×256: 10步采样，不同CFG系数生成样本（参见表12）

ImageNet 256×256: CFG=1.5 vs steps

- OCM tracks or improves FID across steps.
- Recall increases steadily with steps.
- OCM-DDIM reaches strong FID at low steps.
- Consistent improvements over baselines.



Figure: 不同采样步数的生成样本，CFG=1.5，ImageNet 256×256（见表13）

- M=1 already works well.
- Larger M yields marginal visual stability.
- Training remains efficient with M=1.
- Supports practicality of OCM.

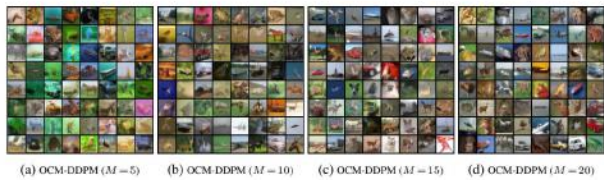


Figure: CIFAR-10 (CS) 上不同Rademacher样本数的10步采样生成样本 (见表1)

- High-quality samples at minimal steps.
- Confirms efficiency without collapsing modes.
- Visuals match cost-adjusted table results.
- Scales across datasets.

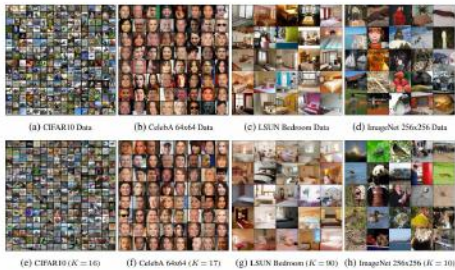


Figure: OCM-DPM在最少步数下达成FID≈6的训练数据与生成样本 (见表4)

Limitations and considerations

- Diagonal covariance only; no off-diagonals.
- Small overhead; up to ~11% in heavy pixels.
- Relies on a strong pretrained score network.
- Schedule and CFG choices still matter.

When to use OCM

- Few-step generation with quality and diversity.
- Tasks needing tractable, calibrated likelihoods.
- Recall-critical applications and coverage.
- Large-scale latent diffusion (e.g., DiT).

- OCM learns optimal diagonal covariance from the score.
- Unbiased objective reduces covariance error.
- Improves FID–NLL, Recall, and sampling efficiency.
- Easy to add: a lightweight Hessian head.

- Structured covariances: low-rank or block-diagonal.
- Integrate with advanced ODE/SDE solvers.
- Scale to video and inverse problems.
- AI4Science, compression, calibrated generative modeling.

- Thank you for your attention!
- Questions and feedback are welcome.

- Thanks to collaborators, institutions, and funding agencies.
- Appreciation to authors of pretrained backbones used.
- Community contributions and open-source tools acknowledged.